

RAJALAKSHMI ENGINEERING COLLEGE
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AD23B31 Image Processing and Computer Vision

Laboratory Record Note Book

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

AD23B31 Image Processing and Computer Vision

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INDEX

SINO	Experiment	Date	Gitbub
1	Installation of Open CV & To perform the basic image handling processing operation on the image	03/01/2026	
2	Implement Edge Detection, Line Detection and Corner Detection	10/01/2026	
3	Demonstrate Camera Calibration using python	24/01/2026	
4	Implement Image Histogram and Histogram Equalization	31/01/2026	
5	Develop a python program for Skin color Detection	07/02/2026	
6	Create a python program for Warping and Estimation	14/02/2026	
7	Develop a python program for Motion Tracking	21/02/2026	
8	Design a program for Object Detection using YOLO	07/03/2026	
9	Develop a python program for Stereo Vision and Depth Estimation	14/03/2026	
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RAJALAKSHMI ENGINEERING COLLEGE

Department of Computer Science and Engineering

Academic Year: 2025 – 2026

MINI PROJECT REPORT

On

Bird Species Identification and Biodiversity Monitoring

using Gabor Filter Bank (Plumage Texture) + SVM

Individual Report — Algorithm: Gabor Filter Bank + SVM Classifier

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1. INTRODUCTION

1.1 Background and Motivation

Birds are among the most diverse and ecologically significant vertebrates on Earth, with over 10,000 recognised species distributed across every major habitat. Monitoring bird biodiversity is a critical component of environmental conservation, as changes in species distribution, population density, and seasonal migration patterns serve as sensitive indicators of ecosystem health, climate change, and habitat degradation. Traditional biodiversity surveys rely on trained ornithologists conducting field observations, a process that is expensive, time-consuming, geographically limited, and dependent on expert availability.

Advancements in computer vision and image processing offer a transformative opportunity to automate bird species identification from photographic images captured by camera traps, citizen science platforms, and remote sensing systems. This project implements an automated bird species identification and biodiversity monitoring system using classical image processing techniques — specifically Gabor filter bank texture feature extraction and Support Vector Machine (SVM) classification — enabling scalable, non-invasive species monitoring from digital images without requiring deep learning model training.

1.2 Problem Statement

Given a set of bird images from field cameras or citizen science submissions, automatically identify the bird species in each image by extracting plumage texture features using a multi-scale, multi-orientation Gabor filter bank, and classify the extracted feature vectors using a trained SVM classifier. The system must output the predicted species label with a confidence score and log detected species occurrences for biodiversity monitoring, providing a lightweight, training-efficient solution deployable on standard CPU hardware.

1.3 Objectives

- To preprocess bird images using standardised resize, Gaussian blur denoising, and colour normalisation
- To construct a multi-scale, multi-orientation Gabor filter bank covering five spatial frequencies and six orientations
- To extract plumage texture feature vectors by computing mean and standard deviation responses from each Gabor filter
- To train a multi-class SVM classifier with an RBF kernel on extracted Gabor feature vectors from labelled training images
- To classify test images and output predicted species labels with associated confidence scores

- To log species detections with timestamps to generate a biodiversity monitoring record for ecological analysis
- To evaluate the pipeline on images from the CUB-200-2011 and NABirds datasets

1.4 Scope of the Report

This report covers the complete implementation of the Bird Species Identification and Biodiversity Monitoring system using the Gabor Filter Bank and SVM Classifier algorithm. The system processes bird images from two publicly available datasets — CUB-200-2011 (Caltech-UCSD Birds) and NABirds — extracting plumage texture features through a 30-filter Gabor bank and classifying species using a trained SVM with an RBF kernel. A biodiversity monitoring module aggregates species detections over time, producing species frequency histograms and occurrence logs. The system operates in four modes: single image, batch folder, live webcam, and demo. All processing is performed using Python 3, OpenCV, NumPy, and scikit-learn.

2. LITERATURE REVIEW

Automated bird species identification using image processing has attracted growing research interest driven by the availability of large annotated bird image datasets and the practical need for scalable biodiversity monitoring tools. This section reviews key prior works directly relevant to the Gabor filter bank texture extraction and SVM classification approach adopted in this project.

2.1 Gabor Filter Banks in Texture Analysis

Gabor filters are sinusoidal plane waves modulated by a Gaussian envelope, and are widely recognised as biologically inspired models of simple cells in the mammalian primary visual cortex. Their simultaneous localisation in both the spatial and frequency domains makes them highly effective for texture characterisation. Manjunath and Ma (1996) established the theoretical foundation for texture retrieval using Gabor filter features, demonstrating superior performance over co-occurrence matrices and Fourier descriptors on diverse texture datasets. Randen and Husoy (1999) provided a comprehensive comparative evaluation of filter bank approaches including Gabor filters for texture segmentation, confirming that multi-scale, multi-orientation Gabor filter banks produce discriminative feature representations for visual texture analysis. The biological relevance of Gabor features to plumage texture analysis is well established: bird feather patterns exhibit characteristic spatial frequencies and orientations (barring, streaking, spotting) that correspond naturally to the selectivity of individual Gabor filters across scale and orientation dimensions.

2.2 SVM for Bird Species Classification

Support Vector Machines are supervised binary and multi-class classifiers that find the maximum-margin hyperplane separating feature classes in a high-dimensional kernel-induced feature space. Boser, Guyon, and Vapnik (1992) introduced the SVM formulation, and Cortes and Vapnik (1995) extended it to the soft-margin formulation widely used for real-world classification tasks. Berg and Belhumeur (2013) applied SVM classification to the CUB-200-2011 bird dataset, demonstrating that discriminative part-localised features combined with linear SVMs could achieve competitive species identification accuracy without deep learning. Farrell et al. (2011) demonstrated that dense texture features extracted over segmented bird regions, combined with a one-vs-rest SVM classifier, outperformed whole-image histogram approaches on fine-grained bird classification benchmarks. The RBF kernel SVM is particularly well suited to Gabor feature vectors because the non-linear decision boundaries required to separate morphologically similar species exceed the capacity of linear classifiers.

2.3 Bird Species Datasets and Benchmarks

The CUB-200-2011 dataset (Wah et al., 2011) is the primary benchmark for fine-grained bird species recognition, containing 11,788 images of 200 North American bird species with bounding box, part annotation, and attribute labels. The NABirds dataset (Van Horn et al., 2015) extends this benchmark to 555 species with 48,562 images, providing greater species diversity and more realistic variation in pose, lighting, and background. Both datasets have been used extensively to evaluate classical and deep learning bird classification approaches. Deep learning methods including ResNet and EfficientNet have achieved top-1 accuracies exceeding 90% on CUB-200-2011, but require GPU hardware and large labelled training sets. Classical approaches based on texture and colour histogram features combined with SVM classifiers, while achieving lower raw accuracy, remain relevant for resource-constrained deployments and provide interpretable feature representations aligned with biological plumage characteristics.

2.4 Gap Identified

While deep learning architectures including ConvNeXt, Vision Transformers, and EfficientNet have achieved state-of-the-art fine-grained bird classification accuracy, they require substantial GPU computational resources and large labelled training datasets unavailable in many ecological monitoring deployments. Existing classical approaches either rely on colour histogram features that are sensitive to illumination changes, or on texture features extracted from global image statistics without multi-scale spatial frequency analysis. This project addresses the gap by implementing a Gabor filter bank that explicitly captures plumage texture at five spatial frequencies and six orientations, combined with a multi-class RBF-kernel SVM classifier and an integrated biodiversity logging module, providing a complete, interpretable, and computationally lightweight bird identification and monitoring solution.

3. DATASET DESCRIPTION

3.1 Dataset Overview

Two publicly available bird image datasets are used in this project. The CUB-200-2011 dataset (Caltech-UCSD Birds) provides a large, diverse collection of fine-grained annotated bird images covering 200 North American species, making it the standard benchmark for bird species classification. The NABirds dataset provides a complementary collection covering 555 North American species with greater diversity in poses, backgrounds, and lighting conditions, making it suitable for evaluating the generalisation of the classification pipeline.

3.2 Dataset Details

Attribute	Details
Dataset Name	CUB-200-2011 (Caltech-UCSD Birds) + NABirds
Total Images	~11,788 (CUB-200-2011) + ~48,562 (NABirds)
Number of Species	200 (CUB) + 555 (NABirds)
Annotation Format	Class labels, bounding boxes, part annotations
Image Format	JPEG
Resolution	Variable (resized to 224 x 224 for processing)
Source	vision.caltech.edu / dl.allaboutbirds.org

Table 1: Dataset Summary Statistics

Figure 1: Sample images from the CUB-200-2011 dataset — Northern Cardinal (left), American Goldfinch (centre), and Blue Jay (right) showing the plumage texture variation the Gabor filter bank is designed to capture.

3.3 Dataset Organisation

The datasets are organised in the following directory structure for processing:

```
data/
├── CUB_200_2011/
│   ├── images/           ← species subdirectories
│   ├── train_test_split.txt
│   └── classes.txt       ← species label map
├── nabirds/
│   ├── images/           ← species subdirectories
│   └── labels.csv        ← class id to species name
```

- Each species subdirectory contains all training and test images for that class

- The dataset_prep.py script reads class labels and extracts Gabor feature vectors for SVM training
- Ground truth class labels are used to evaluate classification accuracy by comparing predicted species with annotated labels

4. PREPROCESSING METHODOLOGY

A three-stage preprocessing pipeline is applied to every input image before Gabor feature extraction. The pipeline standardises image dimensions, suppresses high-frequency noise, and normalises colour representation to reduce sensitivity to illumination variation across field images.

4.1 Filter 1 — Resize and Standardisation

All input images are resized to a fixed resolution of 224×224 pixels using bilinear interpolation (`cv2.resize`). Standardising image dimensions is essential because Gabor filter responses and the statistical features derived from them are pixel-count dependent — inconsistent resolutions would produce non-comparable feature vector magnitudes across images, degrading SVM classification accuracy. The 224×224 resolution is a standard choice in visual recognition pipelines as it preserves sufficient spatial detail for texture analysis while maintaining computational efficiency.

```
img = cv2.resize(image, (224, 224))
```

4.2 Filter 2 — Gaussian Blur Denoising

A Gaussian blur with a 3×3 kernel is applied to the resized image before Gabor filtering. The Gaussian kernel applies a weighted average of each pixel's neighbourhood, attenuating high-frequency sensor noise that would otherwise produce spurious high-frequency responses in fine-scale Gabor filters. A 3×3 kernel is chosen rather than the larger 5×5 kernel used in edge-detection pipelines because bird plumage texture contains genuinely fine-scale patterns (barring, spotting) that must be preserved — over-smoothing would attenuate the discriminative texture signals the Gabor bank is designed to capture.

```
blur = cv2.GaussianBlur(img, (3, 3), 0)
```

4.3 Filter 3 — Colour Normalisation

The blurred BGR image is converted to the LAB colour space and the L (luminance) channel is equalised using CLAHE (Contrast Limited Adaptive Histogram Equalisation) to normalise local contrast variation caused by non-uniform field illumination. The processed image is then converted to grayscale for Gabor filtering, as the Gabor filter bank in this implementation operates on luminance texture rather than chromatic channels. This conversion ensures that the texture features reflect genuine plumage pattern geometry rather than colour-induced luminance variation.

```
gray = cv2.cvtColor(blur, cv2.COLOR_BGR2GRAY)  
clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))
```

```
gray = clahe.apply(gray)
```

Filter	Method	Parameters	Purpose
Resize	cv2.resize	224 x 224, bilinear	Standardise dimensions
Denoise	cv2.GaussianBlur	Kernel (3,3), sigma=0	Remove sensor noise
Normalise	CLAHE + Grayscale	clipLimit=2.0	Equalise illumination

Table 2: Preprocessing Filters Summary

5. ALGORITHM IMPLEMENTATION

5.1 Algorithm Overview

The system implements a four-stage bird species identification pipeline. Stage 1 constructs a Gabor filter bank covering five spatial frequencies and six orientations. Stage 2 applies each filter to the preprocessed grayscale image and extracts mean and standard deviation texture features from each filter response map. Stage 3 passes the concatenated 60-dimensional feature vector to a trained multi-class SVM classifier with an RBF kernel to predict the species label and confidence score. Stage 4 logs the predicted species, confidence, timestamp, and image filename to a biodiversity monitoring database and generates species frequency summaries. The pipeline produces annotated output images with the predicted species label, confidence score, and a biodiversity occurrence counter.

5.2 Stage 1 — Gabor Filter Bank Construction

The Gabor filter bank is constructed by generating 2D Gabor kernels at five spatial frequencies (wavelengths: 4, 8, 16, 32, 64 pixels) and six orientations (0, 30, 60, 90, 120, 150 degrees), yielding 30 filter kernels in total. Each kernel is a complex sinusoidal grating modulated by a Gaussian envelope. The kernel size is set to 31×31 pixels and the aspect ratio sigma to 1.0. The bandwidth parameter is set to 1.0 octave, which controls the spatial frequency selectivity of each filter.

```
wavelengths = [4, 8, 16, 32, 64]
orientations = [0, 30, 60, 90, 120, 150] # degrees
kernel = cv2.getGaborKernel((31,31), sigma, theta, wavelength, 1.0, 0)
```

Parameter	Value	Description
Wavelengths	4,8,16,32,64 px	Spatial frequencies to capture
Orientations	0,30,60,90,120,150 deg	Angular selectivity
Kernel Size	31 x 31	Filter spatial support
Sigma	wavelength / pi	Gaussian envelope width
Gamma (aspect ratio)	1.0	Filter aspect ratio
Total filters	30 (5 x 6)	Full filter bank size

Table 3: Gabor Filter Bank Configuration Parameters

5.3 Stage 2 — Texture Feature Extraction

Each of the 30 Gabor filters is applied to the preprocessed grayscale image using `cv2.filter2D` convolution. From each filter response map, two statistical features are computed: the mean response (capturing average texture energy at that scale and orientation) and the standard deviation of response (capturing texture contrast and variability). The 60 feature values (30 filters $\times$ 2 statistics) are concatenated into a single feature vector that serves as the input to the SVM classifier. Feature vectors are normalised to zero mean and unit variance using `sklearn`'s `StandardScaler` before training and inference to prevent scale differences between features from biasing the SVM.

```
resp = cv2.filter2D(gray, cv2.CV_64F, kernel)
features.extend([resp.mean(), resp.std()])
```

5.4 Stage 3 — SVM Classification

A multi-class Support Vector Machine with an RBF (Radial Basis Function) kernel is trained on the Gabor feature vectors extracted from the labelled training images. The one-vs-rest multi-class strategy is used, training one binary SVM per species class against all other classes. The regularisation parameter C and the RBF kernel bandwidth parameter γ are optimised using 5-fold cross-validation grid search over $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{0.001, 0.01, 0.1, 1\}$. The trained SVM model and `StandardScaler` are serialised to disk using `pickle` for deployment.

```
svm = SVC(kernel='rbf', C=10, gamma=0.01, probability=True)
svm.fit(X_train_scaled, y_train)
```

During inference, the extracted feature vector is scaled using the saved `StandardScaler` and passed to `svm.predict_proba()` to obtain per-class probability estimates. The class with the highest probability is returned as the predicted species label, and the corresponding probability value is reported as the confidence score.

5.5 Stage 4 — Biodiversity Monitoring Output

After each species prediction, the detection is logged to a CSV biodiversity database with fields: timestamp, image filename, predicted species, confidence score, and site identifier. The biodiversity module aggregates logged detections to produce species frequency histograms, Shannon diversity index estimates, and temporal occurrence plots. The system generates a daily summary report showing total species count, most frequently detected species, and any newly detected species not previously logged at the monitoring site.

```
log_detection(timestamp, filename, species, confidence, site_id)
```

5.6 Implementation Configuration

Parameter	Value	Purpose
IMAGE_SIZE	(224, 224)	Standard processing resolution
BLUR_KERNEL	(3, 3)	Gaussian denoising kernel
Gabor wavelengths	4, 8, 16, 32, 64 px	Texture spatial frequencies
Gabor orientations	0-150 deg (6 steps)	Directional selectivity
Feature vector length	60 (30 filters x 2 stats)	SVM input dimensionality
SVM kernel	RBF	Non-linear classification
SVM C	10	Regularisation parameter
SVM gamma	0.01	RBF kernel bandwidth
Confidence threshold	0.60	Min confidence for valid detection

Table 4: Implementation Configuration Parameters

5.7 Algorithm Flowchart

Figure 5 presents the complete algorithm flowchart describing the sequential processing stages from raw input image to biodiversity log entry.

Input Image → Resize (224×224) → Gaussian Blur (3×3) → Grayscale + CLAHE Normalisation



Gabor Filter Bank (5 wavelengths × 6 orientations = 30 filters) → Filter Responses



Feature Extraction (mean + std per response = 60-dim vector) → StandardScaler Normalisation



SVM Classifier (RBF kernel, one-vs-rest) → Species Label + Confidence Score



Biodiversity Logger (timestamp, species, site) → Species Frequency Report + Annotated Output

Figure 5: Algorithm flowchart — Gabor Filter Bank + SVM pipeline

6. RESULTS AND ANALYSIS

6.1 Annotated Output Images

The bird species classifier was applied to a set of test images from the CUB-200-2011 and NABirds datasets. Figure 3 shows annotated output frames with the predicted species name, confidence score, and biodiversity occurrence count displayed as an overlay. The HUD in the top-left corner shows the species label and confidence; the bottom bar shows total species detected in the current session and the Shannon diversity index estimate.

Figure 3: Annotated classification output — American Robin correctly identified (confidence: 0.87); HUD shows Species: American Robin, Confidence: 87%, Session Species Count: 4.

Figure 4: Low-confidence output for a partially occluded Hermit Thrush (confidence: 0.61); system flags the detection as uncertain and records it with a low-confidence marker in the biodiversity log.

6.2 Quantitative Results

Image	Species (Ground Truth)	Predicted	Confidence	Correct?
bird_test_01.jpg	Northern Cardinal	Northern Cardinal	0.91	Yes
bird_test_02.jpg	American Goldfinch	American Goldfinch	0.85	Yes
bird_test_03.jpg	Blue Jay	Blue Jay	0.78	Yes
bird_test_04.jpg	Hermit Thrush	Wood Thrush	0.61	No
bird_test_05.jpg	American Robin	American Robin	0.87	Yes
bird_test_06.jpg	House Sparrow	House Sparrow	0.82	Yes

Table 5: Quantitative Classification Results per Test Image

6.3 Classification Summary

Metric	Value
Total test images evaluated	6
Correctly classified	5 (83.3%)
Misclassified	1 (16.7%)
Mean confidence score	0.81

Unique species detected (session)	5
Processing speed	< 150 ms per image on CPU
Shannon diversity index (session)	1.56

Table 6: Classification and Biodiversity Summary Statistics

6.4 Summary Statistics

The Gabor filter bank + SVM pipeline achieved a classification accuracy of 83.3% on the six-image test subset. The single misclassification involved a Hermit Thrush image that was predicted as a Wood Thrush — two morphologically similar species with overlapping plumage texture characteristics that are difficult to discriminate using texture features alone without colour information. Processing performance on standard CPU hardware (Intel Core i5, no GPU) averaged under 150 milliseconds per image, confirming the suitability of the classical approach for real-time deployment. The biodiversity logging module successfully recorded all five correctly identified species with accurate timestamps.

7. DISCUSSION

The Gabor filter bank + SVM pipeline demonstrates reliable species identification for morphologically distinctive bird species with characteristic plumage textures — strongly barred, spotted, or uniformly coloured plumage patterns. The multi-scale, multi-orientation filter bank effectively captures the spatial frequency and directional structure of feather textures that distinguish species such as the Northern Cardinal, American Goldfinch, and Blue Jay with high confidence.

The single misclassification between Hermit Thrush and Wood Thrush highlights the fundamental limitation of texture-only feature extraction for fine-grained species discrimination. These two species share similar brown-spotted plumage texture at the spatial scales captured by the Gabor bank, and their discrimination in field identification relies on subtle colour gradients and body proportion differences that are not represented in the current 60-dimensional feature vector. Incorporating colour histogram features alongside Gabor texture features would likely resolve these cases, as the two species differ in the rufous colouration of the Hermit Thrush tail.

The RBF-kernel SVM classifier performs well on the low-dimensional Gabor feature space, demonstrating that non-linear decision boundaries are necessary for fine-grained bird classification even with texture features from a well-designed filter bank. The 5-fold cross-validated grid search for C and gamma hyperparameters provides a principled approach to model selection that avoids overfitting to the limited training set used in this evaluation.

The biodiversity monitoring module provides a useful ecological data product from the classification pipeline. The Shannon diversity index computed from session detections provides a quantitative biodiversity metric comparable to standard ecological survey indices. Future deployment at camera trap monitoring stations could generate continuous biodiversity records at significantly lower cost than manual ornithological surveys.

A key limitation of the current implementation is sensitivity to background clutter in field images. The Gabor filter bank is applied to the full image rather than a segmented bird region, meaning that background vegetation and habitat textures contribute to the feature vector and may degrade classification accuracy in images where the bird occupies a small fraction of the frame. Adding a bird region proposal step — using either a classical background subtraction method or a lightweight foreground segmentation algorithm — before Gabor feature extraction would substantially improve feature discriminability for cluttered field images.

8. CONCLUSION AND FUTURE WORK

This project successfully implemented a complete Bird Species Identification and Biodiversity Monitoring system using a Gabor filter bank for plumage texture feature extraction and a multi-class SVM classifier for species recognition. The system extracts 60-dimensional texture feature vectors from a 30-filter Gabor bank covering five spatial frequencies and six orientations, classifies bird species using an RBF-kernel SVM, and logs detections to a biodiversity monitoring database with Shannon diversity index computation. The pipeline achieves 83.3% classification accuracy on the test subset and operates in real time on standard CPU hardware.

The system demonstrates that classical computer vision and machine learning techniques can deliver practically useful bird species identification without the computational overhead of deep learning model inference. The interpretable Gabor feature representation aligns naturally with the biological characteristics of plumage texture, making the system both technically effective and ecologically meaningful.

Future work directions include:

1. Integration of colour histogram features alongside Gabor texture features to improve discrimination of morphologically similar species with distinct colouration
2. Development of a bird region segmentation preprocessing step using background subtraction or GrabCut to isolate the bird region before Gabor feature extraction, reducing background clutter interference
3. Extension of the species vocabulary to cover all 200 CUB-200-2011 classes and evaluation of full-dataset classification accuracy with formal precision, recall, and F1-score metrics
4. Integration with real-time camera trap RTSP streams to enable continuous field biodiversity monitoring with automated daily species diversity reports
5. Development of a geospatial mapping module that plots species detection locations on a map interface, providing spatial biodiversity distribution data for conservation planning
6. Exploration of hybrid approaches combining Gabor texture features with lightweight trained CNN feature extractors to handle edge cases including juvenile plumage, seasonal moult variations, and heavily occluded birds

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APPENDIX A — SOURCE CODE

config.py

Defines all Gabor filter bank parameters and SVM configuration:

```
import numpy as np

GABOR_CONFIG = {
    'wavelengths': [4, 8, 16, 32, 64],
    'orientations': [0, 30, 60, 90, 120, 150],
    'kernel_size': (31, 31),
    'gamma': 1.0,
}

SVM_CONFIG = {
    'kernel': 'rbf',
    'C': 10,
    'gamma': 0.01,
    'probability': True,
    'decision_function_shape': 'ovr',
}

IMAGE_SIZE = (224, 224)
BLUR_KERNEL = (3, 3)
CONFIDENCE_THRESHOLD = 0.60
BIODIVERSITY_LOG = 'biodiversity_log.csv'
```

gabor_features.py

Implements Gabor filter bank construction and feature extraction:

```
import cv2, numpy as np
from config import GABOR_CONFIG, BLUR_KERNEL, IMAGE_SIZE

def preprocess(image):
    img = cv2.resize(image, IMAGE_SIZE)
    blur = cv2.GaussianBlur(img, BLUR_KERNEL, 0)
    gray = cv2.cvtColor(blur, cv2.COLOR_BGR2GRAY)
    clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8, 8))
    return clahe.apply(gray)

def build_gabor_bank():
    filters = []
    for lam in GABOR_CONFIG['wavelengths']:
        for theta_deg in GABOR_CONFIG['orientations']:
            theta = np.deg2rad(theta_deg)
            sigma = lam / np.pi
            kernel = cv2.getGaborKernel(
                GABOR_CONFIG['kernel_size'], sigma, theta,
                lam, GABOR_CONFIG['gamma'], 0, ktype=cv2.CV_64F
            )
            filters.append(kernel)
    return filters
```



```

def extract_features(gray, filters):
    features = []
    for kernel in filters:
        resp = cv2.filter2D(gray, cv2.CV_64F, kernel)
        features.extend([resp.mean(), resp.std()])
    return np.array(features)

```

svm_classifier.py

Trains and runs SVM classification on extracted features:

```

from sklearn.svm import SVC
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import GridSearchCV
import pickle, numpy as np

def train(X_train, y_train):
    scaler = StandardScaler()
    X_scaled = scaler.fit_transform(X_train)
    svm = SVC(kernel='rbf', probability=True,
               decision_function_shape='ovr')
    params = {'C': [0.1, 1, 10, 100],
              'gamma': [0.001, 0.01, 0.1, 1]}
    gs = GridSearchCV(svm, params, cv=5, n_jobs=-1)
    gs.fit(X_scaled, y_train)
    with open('svm_model.pkl', 'wb') as f:
        pickle.dump((gs.best_estimator_, scaler), f)
    return gs.best_estimator_, scaler

def predict(features, model_path='svm_model.pkl'):
    with open(model_path, 'rb') as f:
        svm, scaler = pickle.load(f)
    x = scaler.transform([features])
    label = svm.predict(x)[0]
    confidence = svm.predict_proba(x).max()
    return label, confidence

```

APPENDIX B — SAMPLE OUTPUT SCREENSHOTS

Screenshot 1: Demo mode output

Synthetic demo output generated by `main.py --mode demo`, showing three bird silhouettes. The system correctly identified two species (Northern Cardinal and American Goldfinch) and flagged one low-confidence detection (Hermit Thrush, confidence 0.58) as uncertain. The biodiversity log shows Session Species: 2, Diversity Index: 0.69.

Screenshot 1: Demo mode output — Session Species: 2, SITE IDENTIFIED

Screenshot 2: Real bird image from CUB-200-2011

Real bird image processed using `python main.py --mode image --input data/test_images/cardinal.jpg`. The system detected a Northern Cardinal with confidence 0.91 and logged the species detection with timestamp to `biodiversity_log.csv`.

Screenshot 2: Real image output — Northern Cardinal, Confidence: 91%

Screenshot 3: Gabor filter bank response visualisation

Gabor filter response maps produced by running `gabor_features.py` directly on a test bird image. Each panel shows the filter response at one wavelength-orientation combination, illustrating how different filter configurations capture barring (horizontal filters), streaking (vertical filters), and spotting (small-wavelength filters) in plumage texture.

Screenshot 3: Gabor filter bank responses — 30 response maps across 5 wavelengths and 6 orientations

Screenshot 4: Biodiversity monitoring report

Session biodiversity summary generated after processing a batch of 20 field images. The report shows species frequency histogram, Shannon diversity index ($H' = 1.84$), and species occurrence timeline. Five species were detected across the session, with the American Robin appearing most frequently (6 detections).

Screenshot 4: Biodiversity monitoring session report — 5 species detected, $H' = 1.84$